

Tanuj Haresh Sidhwa

B.Tech in Mechanical Engineering (Minor: Astronomy)

Indian Institute of Technology Indore

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[GitHub](#) | [LinkedIn](#)



EDUCATION

• Indian Institute of Technology, Indore

2023 - 2027

– B.Tech in Mechanical Engineering, CGPA: 9.47/10

Indore, India

– Minor in Astronomy, CGPA: 9.75/10

EXPERIENCE

• Liquid Propulsion Systems Centre (LPSC), ISRO

Jun. 2026 - Jul. 2026

Research Intern / Guide: Vijay Kumar, Scientist/Engineer 'SG'

Bengaluru, India

- Developed and improved an OpenFOAM-based electrohydrodynamic solver for cone-jet electrosprays through boundary-condition corrections, mesh refinement, interface treatment, and storage optimization.
- Validated the computational framework against LPSC experimental cone-jet observations, reproducing stable Taylor-cone formation and physically consistent electrospray behavior for subsequent parametric studies.
- Investigated the effects of voltage, flow rate, electrode spacing, and emitter geometry, identifying electric-field strength and emitter design as primary drivers of emitted current and electrospray performance.

• IASc-INSa-NASI Summer Research Fellowship

May 2025 - Jul. 2025

Research Intern / Dr Abhijit Kushari, Advanced Propulsion Laboratory

Indian Institute of Technology, Kanpur

- Conducted static structural analysis of serpentine nozzles in ANSYS, evaluating deformation behavior under fluid-induced and inertial loading conditions relevant to propulsion and high-speed flow applications.
- Performed fatigue, modal, and harmonic analyses to estimate component lifespan, identify resonance conditions through Campbell diagrams, and evaluate structural integrity under complex dynamic excitation environments.
- Studied plasticity model implementation using bilinear isotropic hardening models, while developing proficiency in CATIA V5 and conducting preliminary fluid-structure interaction studies supported by literature review.

PROJECTS

• Powered Lift Wing Design for STOL Applications

Oct. 2025 - Dec. 2025

InterIIT Tech Meet 14.0: High Prep – LAT Aerospace

- Led a multidisciplinary team in designing a powered-lift STOL wing for InterIIT Tech Meet 14.0, coordinating aerodynamics, propulsion, and stability efforts while managing system-level integration and technical workflows.
- Directed CFD-driven design iterations in ANSYS Fluent and Workbench, overseeing airfoil selection, high-lift configurations, and multi-fidelity aerodynamic modeling through analytical validation and parametric studies.
- Optimized the wing design through trade-off studies balancing lift enhancement, drag, stability, and computational cost, achieving a maximum lift coefficient (C_L) of 6.81 and documenting results in a comprehensive technical report.

• Assessment of Indoor Air Quality in Student Hostel

Sept. 2024 - Oct. 2025

Prof. Ritunesh Kumar, Department of Mechanical Engineering

[GitHub](#)

- Developed an Arduino-based indoor air quality monitoring system using MH-Z19B (NDIR CO₂) and DHT11 sensors to continuously record CO₂ concentration and environmental data inside student hostel rooms.
- Conducted controlled occupancy and ventilation experiments to analyze time-dependent CO₂ buildup trends; resulting work led to a first-author paper accepted at INCOTHERM 2025, IIT(ISM) Dhanbad.

• Physics-Informed Neural Network for Engine Thermodynamics

Jan. 2026 - Present

Prof. Devendra Deshmukh, Department of Mechanical Engineering

- Developed a Pointwise Fourier Physics-Informed Neural Network (PINN) in PyTorch to predict instantaneous in-cylinder pressure using high-fidelity diesel combustion data from the Sandia Engine Combustion Network (ECN).
- Integrated the First Law of Thermodynamics into a custom loss function with dynamic specific heat ratio modeling, and mitigated gradient conflicts and Fourier leakage using L1 Total Variation regularization, reducing inference error on unseen injection profiles to below 7%.

• Numerical Design and Performance Mapping of a Micro Cold-Gas Thruster

Jan. 2026 - Apr. 2026

Course Project: Computational FLuids and Structures (AA374), IIT Indore

[GitHub](#)

- Developed a quasi-one-dimensional finite volume solver for compressible Euler equations to model converging-diverging nozzle flows, capturing transonic acceleration, shock formation, and supersonic expansion.
- Implemented and compared Lax–Friedrichs, HLLC, and MUSCL–HLLC schemes with TVD limiters, validating results against analytical isentropic relations and ANSYS Fluent simulations while performing thrust and specific impulse trade-off analysis for micro-thruster design.

• CubeSat Project

Oct. 2024 - Nov. 2024

CAE Club & Astronomy Club, IIT Indore

[GitHub](#)

- Led the CubeSat project team by coordinating workflows, managing cross-domain collaboration, and handling official technical documentation including proposals and reports.
- Designed the primary CubeSat chassis and performed modal, thermal, and structural analyses with detailed load calculations to validate mechanical integrity, thermal resilience, and compliance with aerospace design standards under launch and operational conditions.

TECHNICAL SKILLS

- **Simulation & Aerodynamic Analysis:** ANSYS Mechanical (Static & Transient Structural, Modal, Harmonic, Thermal), ANSYS Fluent*, XFLR5, XFOIL
- **CAD & Design:** CATIA V5, Fusion 360, ANSYS SpaceClaim
- **Scientific Computing & Machine Learning:** Python, MATLAB, C++, NumPy*, SciPy*, Pandas, PyTorch*

* Elementary proficiency

CONFERENCES

• 1st International Conference on Thermofluids Engineering (INCOTHERM-2025)

Oct. 2025

First Author

IIT (ISM) Dhanbad

- First author of the paper presented at this conference, titled “*Experimental Investigations on Effects of Student Occupancy Rate in a Student Hostel Room on Indoor Air Quality*”.

• Students’ Academic Conference (SAC)

Dec. 2025

Presenter

Inter IIT Tech Meet 14.0, IIT Patna

- Presented the paper titled “*An Innovative Laser Decal Transfer of ZnO Ceramic in LIG for Advanced Hybrid Nanogenerator Applications*” on behalf of the primary author.

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Multivariable Calculus, Differential Equations, Real Analysis, Complex Analysis, Numerical Methods
- **Mechanical Core:** Fluid Mechanics, Thermodynamics, Heat Transfer, Applied Thermodynamics, Strength of Materials, Solid Mechanics, Engineering Mechanics, Design of Machine Elements, Kinematics & Dynamics of Machines, Instrumentation and Control Systems, Principles of Industrial Engineering
- **Electives:** Fundamentals of Vibrations, Fundamentals of Acoustics, Introduction to Robotics, Automobile Engineering, Introduction to Electric Vehicles, Computational Fluids and Structures
- **Minor:** Introduction to Astronomy, Flight mechanics and classical control, IoT for Space Applications, High Energy Astrophysics and Transient Sky
- **Certificates:** Rocket Propulsion and Spacecraft Dynamics, Aerospace Engineering: Aircraft Fundamentals and Advanced

ACHIEVEMENTS

- **Top 1 Percentile in JEE 2023**, Ranked in the top 1% among nearly 1 million candidates 2023
- **Top 0.1 Percentile in MHT-CET**, Ranked in the top 0.1% among 6 lakh candidates 2023
- **Reliance Foundation Undergraduate Scholar**, Awarded to meritorious students nationwide 2024
- **Bronze Medal: IITISoC’24**, Design & Modelling, Material Selection, Design and FEA of CubeSat 2024
- **6th Position:** InterIIT Techmeet 14.0, Lat Aerospace HighPrep PS, Powered Lift Wing Design 2025
- **AIR 1224:** GATE Mechanical Engineering (ME), Secured All India Rank 1224 in GATE ME 2026 2026
- **AIR 144:** GATE Aerospace Engineering (AE), Secured All India Rank 144 in GATE AE 2026 2026
- **Awarded AP grade in AA374: Computational Fluids and Structures**, for exceptional performance, securing the top rank in the class. 2026

POSITIONS OF RESPONSIBILITY

- **President**, Mechanical Engineering Student Association, IIT Indore *Apr. 2025 - Apr. 2026*
 - **Team Lead**, High Prep LAT Aerospace, InterIIT Techmeet 14.0 *Oct. 2025 - Dec. 2025*
 - **Project Lead**, CubeSat Project, IIT Indore *Oct. 2024 - Nov. 2024*
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